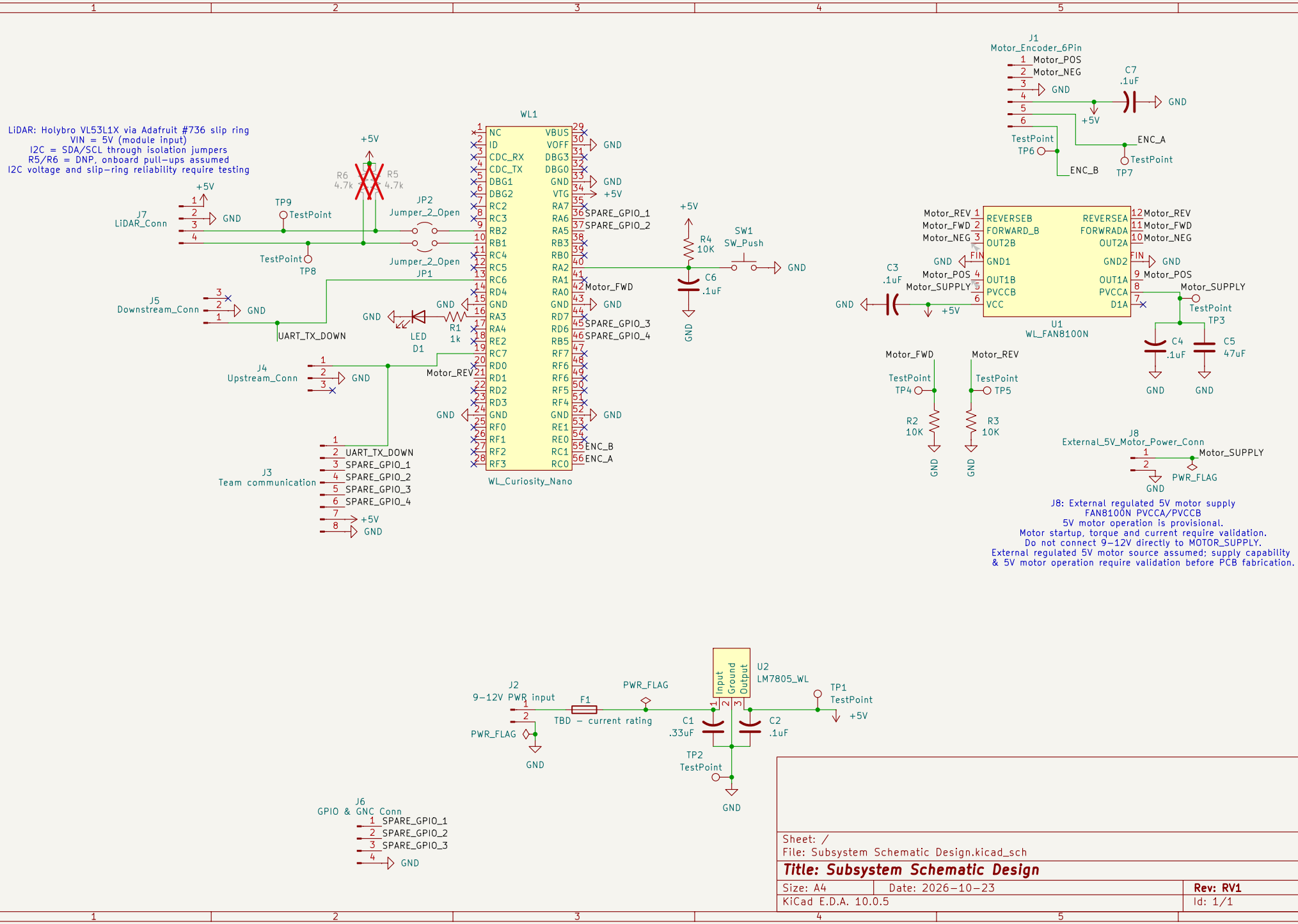
LiDAR: Holybro VL53L1X via Adafruit #736 slip ring
VIN = 5V (module input)
I2C = SDA/SCL through isolation jumpers
R5/R6 = DNP, onboard pull-ups assumed
I2C voltage and slip-ring reliability require testing
WL1
WL_Curiosity_Nano
J1
Motor_Encoder_6Pin
Motor_POS
Motor_NEG
ENC_A
ENC_B
J2
9-12V PWR input
F1
TBD - current rating
PWR_FLAG
U2
LM7805_WL
Input
Ground
Output
J3
Team communication
J4
Upstream_Conn
J5
Downstream_Conn
J6
GPIO & GNC Conn
J7
LiDAR_Conn
J8
External_5V_Motor_Power_Conn
U1
WL_FAN8100N
REVERSEB
FORWARD_B
OUT2B
GND1
OUT1B
PVCCB
VCC
REVERSEA
FORWRADA
OUT2A
GND2
OUT1A
PVCCA
D1A
J8: External regulated 5V motor supply
FAN8100N PVCCA/PVCCB
5V motor operation is provisional.
Motor startup, torque and current require validation.
Do not connect 9-12V directly to MOTOR_SUPPLY.
External regulated 5V motor source assumed; supply capability
& 5V motor operation require validation before PCB fabrication.
Sheet: /
File: Subsystem Schematic Design.kicad_sch
Title: Subsystem Schematic Design
Size: A4
Date: 2026-10-23
Rev: RV1
KiCad E.D.A. 10.0.5
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